



# KTU NOTES

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Module - I

Partial differential Equations.

A differential equation involving more than one independent variable, i.e. partial derivatives is called partial differential equations. (p.d.e). Order of p.d.e is the order of the highest derivative occurring in p.d.e. Degree of the p.d.e is the power of the highest order partial derivative involved in the equation.

For example

$$\textcircled{1} \quad \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} = 0 \quad \left\{ \begin{array}{l} \text{order} - 1 \\ \text{degree} - 1 \end{array} \right\}.$$

$$\textcircled{2} \quad \frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial y} = 0 \quad \left\{ \begin{array}{l} \text{order} - 2 \\ \text{degree} - 1 \end{array} \right\}.$$

$$\textcircled{3} \quad \left( \frac{\partial u}{\partial x} \right)^2 + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^3 u}{\partial z^3} \quad \left\{ \begin{array}{l} \text{order} - 3 \\ \text{degree} - 1 \end{array} \right\}.$$

dependent variable is  $u$ .

independent variables are  $x, y$  and  $z$ .

i.e.  $u$  is function of  $x, y$  and  $z$ .

(2)

$$\textcircled{4} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

Here  $u$  is a function of  $x$  and  $y$ .  
 $x$  and  $y$  are independent variables.

Formation of partial differential equations.

There are two methods for formation

of p.d.e

① Elimination of arbitrary constants.

② Elimination of arbitrary function.

① Form the p.d.e by eliminating arbitrary constants  $a$  and  $b$  from  $z = ax + by + ab$ .

$$z = ax + by + ab. \text{---} \textcircled{1}$$

Take the partial derivative w.r. to  $x$ .

$$\frac{\partial z}{\partial x} = a$$

Take the partial derivative w.r. to  $y$

$$\frac{\partial z}{\partial y} = b.$$

Substitute in ①.

$$z = \left(\frac{\partial z}{\partial x}\right)x + \left(\frac{\partial z}{\partial y}\right)y + \left(\frac{\partial z}{\partial x}\right)\left(\frac{\partial z}{\partial y}\right)$$

This is the required p.d.e

② Form the p.d.e from  $z = a^2x + b^2y + ab$ .

$$\frac{\partial z}{\partial x} = a^2, \quad \frac{\partial z}{\partial y} = b^2$$

Substituting

$$z = \left(\frac{\partial z}{\partial x}\right)x + \left(\frac{\partial z}{\partial y}\right)y + \left(\frac{\partial z}{\partial x}\right)^{1/2} \left(\frac{\partial z}{\partial y}\right)^{1/2}$$

③ Form the p.d.e from the following

(a)  $z = (x-a)^2 + (y-b)^2$

(b)  $z = a(x+y) + b(x-y) + abt + c$

(c)  $z = a e^{bx} \sin by$

Ans: (a) ~~z~~  $z = (x-a)^2 + (y-b)^2$  — ①

$$\frac{\partial z}{\partial x} = 2(x-a) \Rightarrow x-a = \frac{1}{2} \frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial y} = 2(y-b) \Rightarrow y-b = \frac{1}{2} \frac{\partial z}{\partial y}$$

Substituting in ①

$$z = \left(\frac{1}{2} \frac{\partial z}{\partial x}\right)^2 + \left(\frac{1}{2} \left(\frac{\partial z}{\partial y}\right)^2\right)$$

$$= \frac{1}{4} \left(\frac{\partial z}{\partial x}\right)^2 + \frac{1}{4} \left(\frac{\partial z}{\partial y}\right)^2$$

$$4z = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 //$$

Note

Notations are  $\frac{\partial z}{\partial x} = p$ ,  $\frac{\partial z}{\partial y} = q$

$$\frac{\partial^2 z}{\partial x^2} = r, \quad \frac{\partial^2 z}{\partial x \partial y} = s, \quad \frac{\partial^2 z}{\partial y^2} = t$$

(b)  $z = a(x+y) + b(x-y) + abt + c$

$$\frac{\partial z}{\partial x} = a + b. \quad \text{ie } p = a + b$$

$$\frac{\partial z}{\partial y} = a - b.$$

$$\frac{\partial z}{\partial t} = ab.$$

$$(a+b)^2 - (a-b)^2 = 4ab.$$

$$\text{ie } \left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = 4 \frac{\partial z}{\partial t}.$$

This is the required o.d.e

(c)  $z = a e^{bx} \sin by$

$$\frac{\partial z}{\partial x} = a e^{bx} \times b \sin by \Rightarrow b z. \quad \therefore b = \frac{1}{z} \frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial y} = a e^{bx} \cos by \times b.$$

$$\frac{\frac{\partial z}{\partial x}}{\frac{\partial z}{\partial y}} = \frac{a e^{bx} b \sin by}{a e^{bx} \cos by \times b} = \tan by$$

④  $\frac{p}{q} = \tan \theta \Rightarrow p = q \tan \theta$   
 which is the required  $p \cdot d.e. = q \tan \left[ \frac{y}{z} \frac{\partial z}{\partial x} \right]$   
 $p = q \tan \left[ \frac{y}{z} \right]$

④ Form the p.d.e from  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ .  
 partial derivative w.r. to  $x$ .

$$\frac{2x}{a^2} + \frac{2z}{c^2} \frac{\partial z}{\partial x} = 0.$$

$$2c^2 x + 2za^2 \frac{\partial z}{\partial x} = 0.$$

$$c^2 x + za^2 \frac{\partial z}{\partial x} = 0. \text{ --- (1)}$$

$$\text{ie } \frac{c^2}{a^2} = -\frac{z}{x} \frac{\partial z}{\partial x} \text{ --- (2)}$$

Differentiate (1) w.r. to  $x$ .

$$c^2 + a^2 \left[ z \frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial x} \frac{\partial z}{\partial x} \right] = 0.$$

$$c^2 + a^2 \left[ z \frac{\partial^2 z}{\partial x^2} + \left( \frac{\partial z}{\partial x} \right)^2 \right] = 0.$$

$$\frac{c^2}{a^2} = -z \frac{\partial^2 z}{\partial x^2} - \left( \frac{\partial z}{\partial x} \right)^2 \text{ --- (3)}$$

From (2) and (3)

$$-\frac{z}{x} \frac{\partial z}{\partial x} = -z \frac{\partial^2 z}{\partial x^2} - \left( \frac{\partial z}{\partial x} \right)^2$$

$$z \frac{\partial^2 z}{\partial x^2} + \left( \frac{\partial z}{\partial x} \right)^2 - \frac{z}{x} \frac{\partial z}{\partial x} = 0$$

$$zx + p^2 - \frac{z}{x} p = 0 //$$

⑤ Form the p.d.e of all spheres of radius one and centre lies on xy-plane.  
on xy plane —  $z=0$

∴ centre is  $(a, b, 0)$ .  $r=1$ .

$$\therefore (x-a)^2 + (y-b)^2 + (z-0)^2 = 1. \quad \text{--- (1)}$$

Differentiate (1) w.r. to  $x$ .

$$2(x-a) + 2z \frac{\partial z}{\partial x} = 0$$

$$\therefore x-a = -z \frac{\partial z}{\partial x}.$$

Diff. (1) w.r. to  $y$

$$2(y-b) + 2z \frac{\partial z}{\partial y} = 0$$

$$y-b = -z \frac{\partial z}{\partial y}.$$

Substitute  $(x-a)$  and  $(y-b)$  in (1)

$$\left(-z \frac{\partial z}{\partial x}\right)^2 + \left(-z \frac{\partial z}{\partial y}\right)^2 + z^2 = 1.$$

$$z^2 \left[ \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1 \right] = 1.$$

$$\underline{\underline{z^2 [p^2 + q^2 + 1] = 1. \text{ required p.d.e.}}}$$

I Form the p.d.e from

①  $z = (x+a)(y+b)$

②  $z = ax + b^2y^2 + a$

③  $z = (x+a^2)(y+b^2)$

II) Form the p.d.e of all spheres whose centres lies on the  $z$ -axis.

Note.

① If the number of arbitrary constants to be eliminated is equal to the number of independent variables, then the p.d.e that formed is of first order.

② If the number of arbitrary constants to be eliminated is more than the number of independent variables then the p.d.e formed is of second order or higher order.

## Formation of p.d.e by elimination of arbitrary function.

① Form the p.d.e by eliminating the arbitrary function from  $z = f(x^2 + y^2)$

$$z = f(x^2 + y^2)$$

$$\frac{\partial z}{\partial x} = f'(x^2 + y^2) \times 2x. \quad \text{--- (1)}$$

$$\frac{\partial z}{\partial y} = f'(x^2 + y^2) \times 2y \quad \text{--- (2)}$$

Equate  $f'(x^2 + y^2)$  from (1) and (2)  
we get

$$\cancel{\frac{2x}{\partial z / \partial x}} \times \frac{1}{2x} \frac{\partial z}{\partial x} = \frac{1}{2y} \frac{\partial z}{\partial y}.$$

$$x \frac{\partial z}{\partial y} - y \frac{\partial z}{\partial x} = 0.$$

$$\underline{xq - yp = 0} \quad \text{required p.d.e.}$$

② Form the p.d.e from  $xyz = \phi(x+y+z)$

Diff. partially w.r. to  $x$ .

$$y \left[ x \frac{\partial z}{\partial x} + z \right] = \phi'(x+y+z) \left( 1 + \frac{\partial z}{\partial x} \right) \quad \text{--- (1)}$$

Diff. w.r.t. to  $y$

$$x \left[ y \frac{\partial z}{\partial y} + z \right] = \phi'(x+y+z) \left[ 1 + \frac{\partial z}{\partial y} \right] \quad \text{--- (8)}$$

$$\textcircled{1} \Rightarrow \phi'(x+y+z) = \left[ \frac{1}{1 + \frac{\partial z}{\partial x}} \right] \left[ y \left( x \frac{\partial z}{\partial x} + z \right) \right]$$

$$\phi'(x+y+z) = \frac{1}{1+p} \left[ y(xp+z) \right]$$

$$\textcircled{2} \Rightarrow \phi'(x+y+z) = \frac{1}{1+q} \left[ x(yq+z) \right]$$

$$\therefore \left[ \frac{1}{1+p} \right] \left[ y(xp+z) \right] = \frac{1}{1+q} \left[ x(yq+z) \right]$$

③ Form the p.d.e from  $z = f\left(\frac{xy}{z}\right)$

$$z = f\left(\frac{xy}{z}\right)$$

Diff. w.r.t. to  $x$

$$\frac{\partial z}{\partial x} = f'\left(\frac{xy}{z}\right) \left[ \frac{zy - xy \frac{\partial z}{\partial x}}{z^2} \right]$$

$$p = f'\left(\frac{xy}{z}\right) \left[ \frac{zy - xy p}{z^2} \right] \quad \text{--- (1)}$$

Diff. w.r.t. to  $y$

$$\frac{\partial z}{\partial y} = f'\left(\frac{xy}{z}\right) \left[ \frac{zx - xy \frac{\partial z}{\partial y}}{z^2} \right]$$

$$q = f'\left(\frac{xy}{z}\right) \left[ \frac{zx - xy q}{z^2} \right] \quad \text{--- (2)}$$

① ÷ ②

$$\frac{p}{q} = \frac{\frac{zy - xyp}{z^2}}{\frac{zx - xyp}{z^2}} = \frac{zy - xyp}{zx - xyp} \times \frac{z^2}{z^2}$$

$$\therefore \frac{p}{q} = \frac{y(z - xp)}{x(z - yp)}$$

$$px(z - yp) = qy(z - xp)$$

$$pxz - pyqz = zqy - zyxp$$

$$pxz - pyqz - zqy + zyxp = 0$$

$$z[px - qy] = 0$$

$$z \neq 0 \Rightarrow px - qy = 0 \text{ is the}$$

required p.d.e.

④ Form the p.d.e. by eliminating arbitrary function  $z = y f(x) + x g(y)$ .

$$z = y f(x) + x g(y)$$

Diff. w.r. to  $x$ .

$$\frac{\partial z}{\partial x} = y f'(x) + g(y)$$

Diff. w.r. to  $y$

$$\frac{\partial z}{\partial y} = f(x) + x g'(y)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) \\ = f'(x) + g'(y).$$

Consider  $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$

$$= xy f'(x) + x g(y) + y f(x) + yx g'(y) \\ = xy [f'(x) + g'(y)] + x g(y) + y f(x) \\ = xy \frac{\partial^2 z}{\partial x \partial y} + z.$$

$$\therefore x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} - xy \frac{\partial^2 z}{\partial x \partial y} = z$$

is the required p.d.e

⑤ Form the p.d.e  $z = f(x) + e^y g(x).$

$$\frac{\partial z}{\partial x} = f'(x) + e^y g'(x). \quad \text{--- (1)}$$

$$\frac{\partial z}{\partial y} = e^y g(x). \quad \text{--- (2)}$$

Diff. (2) w.r. to  $y$

$$\frac{\partial^2 z}{\partial y^2} = e^y g(x) \quad \text{--- (3)}$$

From (2) and (3)  $\Rightarrow \frac{\partial z}{\partial y} = \frac{\partial^2 z}{\partial y^2}$   
 $q = p \Rightarrow q - p = 0$

⑥ Form the p.d.e from  $f(x^2+y^2, z-xy)=0$ .

We can write the above equation as

$$x^2+y^2 = f(z-xy)$$

$$\text{OR } z-xy = f(x^2+y^2)$$

$$\therefore \text{Consider } z-xy = f(x^2+y^2) \quad \text{--- (1)}$$

Diff. (1) w.r. to  $x$ .

$$\frac{\partial z}{\partial x} - y = f'(x^2+y^2) \cdot 2x \quad \text{--- (2)}$$

Diff. (1) w.r. to  $y$

$$\frac{\partial z}{\partial y} - x = f'(x^2+y^2) \cdot 2y \quad \text{--- (3)}$$

$$\text{(2)} \Rightarrow f'(x^2+y^2) = \frac{1}{2x} \left[ \frac{\partial z}{\partial x} - y \right]$$

$$\text{(3)} \Rightarrow f'(x^2+y^2) = \frac{1}{2y} \left[ \frac{\partial z}{\partial y} - x \right]$$

$$\therefore \frac{1}{2x} \left[ \frac{\partial z}{\partial x} - y \right] = \frac{1}{2y} \left[ \frac{\partial z}{\partial y} - x \right]$$

$$\frac{1}{x} [p - y] = \frac{1}{y} [q - x]$$

$$y(p - y) = x(q - x)$$

$$yp - y^2 = xq - x^2 \Rightarrow \underline{\underline{x^2 - y^2 + yp - xq = 0}}$$

If  $f(u, v) = 0$   
It can be written  
as  $u = f(v)$  or  
 $v = f(u)$ .

H.w

- ① Form the p.d.e by eliminating arbitrary function
- (a)  $f(x+y+z, xy+z^2) = 0$ .
- (b)  $x+y+z = \phi(x^2+y^2+z^2)$ .
- (c)  $z = x f(ax+by) + g(ax+by)$

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Lagrange's linear equations

A linear partial differential equation of the first order, commonly known as Lagrange's linear equations is of the form

$$P \frac{\partial z}{\partial x} + Q \frac{\partial z}{\partial y} = R$$

ie  $Pp + Qq = R$  <sup>①</sup>, where  $P, Q$  and  $R$  are functions of  $x, y$  and  $z$ .

To solve the equation ①, Consider the subsidiary equations,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \text{ --- ②}$$

$U(x, y, z) = a$  and  $V(x, y, z) = b$  are any two independent solutions of the above subsidiary equation ②

Then the complete solution of ① is

$$f(u, v) = 0 \quad \text{or} \quad u = f(v) \\ \text{or} \quad v = f(u).$$

Following methods are used for solving  
Subsidiary equations.

Method-1 : Method of grouping.

$$S.E \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

we can eliminate  $z$  from first two fractions of S.E, then a solution  $u(x, y) = a$  can be obtained by direct integration. Apply the same procedure for the other two pairs of fractions we get the second independent solution.

eg:  $\triangleright$  Solve  $y^2 z \left( \frac{\partial z}{\partial x} + x^2 z \frac{\partial z}{\partial y} \right) = xy^2$

$$\text{ie } y^2 z p + x^2 z q = R.$$

$$P = y^2 z, \quad Q = x^2 z, \quad R = xy^2$$

Form the subsidiary equation.

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

$$\frac{dx}{y^2 z} = \frac{dy}{x^2 z} = \frac{dz}{xy^2} \quad (S.E)$$

Grouping first two fractions.

$$\text{ie } \frac{dx}{y^2 z} = \frac{dy}{x^2 z}$$

$$\Rightarrow \frac{dx}{y^2} = \frac{dy}{x^2}$$

$$\text{ie } x^2 dx = y^2 dy$$

Integrate both sides.

$$\int x^2 dx = \int y^2 dy$$

$$\frac{x^3}{3} = \frac{y^3}{3} + C_1$$

$$x^3 - y^3 = 3C_1 \\ = a \text{ (say)}$$

$$x^3 - y^3 = a \quad \text{--- (1)} \quad \{u=a\}.$$

Grouping first and last fractions in S.E

$$\frac{dx}{y^2 z} = \frac{dz}{xy^2}$$

$$\Rightarrow \frac{dx}{z} = \frac{dz}{x}$$

$$x dx = z dz$$

$$\text{Integrating } \frac{x^2}{2} = \frac{z^2}{2} + C_2$$

$$x^2 - z^2 = 2C_2 = b \quad \text{--- (2)} \quad \{v=b\}$$

(1) and (2) are two independent solutions of subsidiary equation.

∴ Solution of the given Lagrange's equation

is  $f(u, v) = 0$

ie  $f(x^3 - y^3, x^2 - z^2) = 0$

or  $x^3 - y^3 = f(x^2 - z^2)$

or  $x^2 - z^2 = f(x^3 - y^3)$

② Solve  $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = z^2$

ie  $x^2 p + y^2 q = z^2$

$P = x^2$     $Q = y^2$  ,    $R = z^2$

S.E  $\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{z^2}$$

Grouping first and second fractions

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

Integrating  $\int \frac{dx}{x^2} = \int \frac{dy}{y^2}$

$$-\frac{1}{x} = -\frac{1}{y} + a.$$

$$\frac{1}{y} - \frac{1}{x} = a \quad \text{--- ①}$$

Grouping last two fractions,

$$\frac{dy}{y^2} = \frac{dz}{z^2}$$

$$\text{Integrating } -\frac{1}{y} = -\frac{1}{z} + b.$$

$$\text{ie } \frac{1}{z} - \frac{1}{y} = b. \quad \text{--- (2)}$$

① and ② are ~~linearly~~ independent solutions of S.E

$$\therefore \text{Solution is } f\left(-\frac{1}{x} + \frac{1}{y}, \frac{1}{z} - \frac{1}{y}\right) = 0$$

$$\text{or } -\frac{1}{x} + \frac{1}{y} = f\left(\frac{1}{z} - \frac{1}{y}\right)$$

$$\text{or } \frac{1}{z} - \frac{1}{y} = f\left(-\frac{1}{x} + \frac{1}{y}\right)$$

③ Solve  $zp = -x$ . ( $Pp + Qq = R$ )

$$P = z, \quad Q = 0, \quad R = -x.$$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{z} = \frac{dy}{0} = \frac{dz}{-x}$$

Grouping first two fractions,

$$\frac{dx}{z} = \frac{dy}{0} \Rightarrow z dy = 0$$

$$z \neq 0 \Rightarrow dy = 0$$

Grouping first and last

$$\frac{dx}{z} = \frac{dz}{-x} \Rightarrow x dz = -x dx$$

$$y = C_1 \text{ (Const)} \quad \text{--- (1)}$$

Integrating  $\frac{z^2}{2} = -\frac{x^2}{2} + C_2$

$$\frac{x^2}{2} + \frac{z^2}{2} = C_2$$

$$x^2 + z^2 = 2C_2 = b \cdot (\sec y) \quad \text{--- (2)}$$

$\therefore$  Solution is  $f(y, x^2 + z^2) = 0$ .

(4)  $P - Q = \log(x+y)$

$P = 1, Q = -1, R = \log(x+y)$

S.E  $\Rightarrow \frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{\log(x+y)}$

Comparing first two fractions.

$$\frac{dx}{1} = \frac{dy}{-1} \Rightarrow -dx = dy$$

Integrating  $-x = y + a$

$x + y = a \quad \text{--- (1)}$

$$\frac{dx}{1} = \frac{dz}{\log(x+y)}$$

$$\Rightarrow \frac{dx}{1} = \frac{dz}{\log a} \quad (\text{using (1)})$$

$$dx = \frac{1}{\log a} dz$$

Integrating  $x = \frac{z}{\log a} + b$ .

$$\text{ie } x = \frac{z}{\log(x+y)} + b.$$

$$\text{ie } x - \frac{z}{\log(x+y)} = b. \text{---(2)}$$

① and ② are independent solution of subsidiary Equations.

$$\therefore \text{Solution is } \underline{f\left(x+y, x - \frac{z}{\log(x+y)}\right) = 0}$$

$$\textcircled{5} \quad xp + yq = 3z. \quad \{Pp + Qq = R\}.$$

$$P = x, \quad Q = y, \quad R = 3z.$$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{3z}$$

From first two fractions,

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating we get  $\log x = \log y + \log a$   
(Take the constant also  $\log a$ )

$$\therefore \log x - \log y = \log a.$$

$$\log\left(\frac{x}{y}\right) = \log a.$$

$$\text{ie } \frac{x}{y} = a \text{ --- (1). } \{u=a\}$$

$$\text{Take last two fractions, } \frac{dy}{y} = \frac{dz}{3z}$$

Integrating  $\log y = \frac{1}{3} \log z + \log b.$

$$\log y - \frac{1}{3} \log z = \log b.$$

$$\log \left\{ \frac{y}{z^{1/3}} \right\} = \log b.$$

$$\text{ie } \frac{y}{z^{1/3}} = b. \text{ --- (2) } \{v=b\}$$

$\therefore$  (1) and (2) are two independent solutions of subsidiary equation.

$\therefore$  solution is

$$\underline{\underline{\frac{x}{y} = f\left(\frac{y}{z^{1/3}}\right)}}$$

$$\left\{ \begin{array}{l} \text{Solution} \\ f(u, v) = 0 \\ \text{or} \\ u = f(v) \\ \text{or } v = f(u) \end{array} \right\}$$

(6)  $a(p+q) = z \implies ap + aq = z$

$$P=a, Q=a, R=z.$$

$$\frac{dx}{a} = \frac{dy}{a} = \frac{dz}{z}$$

$$\frac{dx}{a} = \frac{dy}{a} \implies dx = dy$$

$$\implies x = y + a$$

$$x - y = a \text{ --- (1)}$$

$$\frac{dy}{a} = \frac{dz}{z}$$

$$\text{Integrating } \frac{1}{a} y = \log z + b.$$

$$\frac{y}{a} - \log z = b \quad \text{--- (2)}$$

(5)

$\therefore$  Solution is

$$f(x-y, \frac{y}{a} - \log z) = 0$$

Method 2 : Method of Multipliers.

Choose any three multipliers  $l, m, n$  (which may be constant or functions of  $x, y, z$ )

Such that in

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{lP + mQ + nR},$$

the expression  $lP + mQ + nR = 0$ ,

then we get  $l dx + m dy + n dz = 0$ .

Integrating we get first solution  $u(x, y, z) = a$

Another independent solution  $v(x, y, z) = b$  can be found by choosing another set of multipliers  $l', m', n'$  such that  $l'P + m'Q + n'R = 0$ .

① Solve  $(mz - ny) \frac{\partial z}{\partial x} + (nx - lz) \frac{\partial z}{\partial y} = ly - mx$

$$P = mz - ny, \quad Q = nx - lz, \quad R = ly - mx$$

$$S.E \Rightarrow \frac{dx}{xz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx}$$

Choose the multiplier  $x, y, z$ .

$$\text{ie } \frac{dx}{xz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{x dx + y dy + z dz}{0}$$

$$\begin{cases} x(xz-ny) + y(nx-lz) + z(ly-mx) \\ = x^2z - xny + ynx - ylz + zly - mxz \\ = 0 \end{cases}$$

Take first and fourth fraction

$$\frac{dx}{xz-ny} = \frac{x dx + y dy + z dz}{0}$$

$$\text{ie } (xz-ny)(x dx + y dy + z dz) = 0$$

$$xz-ny \neq 0 \Rightarrow x dx + y dy + z dz = 0$$

$$\text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = a$$

$$\underline{x^2 + y^2 + z^2 = 2a = c_1} \quad \text{--- (1)}$$

② To find the second independent solution choose the multipliers  $l, m, n$ .

$$\text{ie } \frac{dx}{xz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{l dx + m dy + n dz}{l(xz-ny) + m(nx-lz) + n(ly-mx)} \Rightarrow 0$$

Take the first and last fractions

⑥

$$\frac{dx}{xz-ny} = \frac{l dx + m dy + n dz}{0}$$

$$\text{ie } (xz-ny)(l dx + m dy + n dz) = 0$$

$$\text{ie } l dx + m dy + n dz = 0$$

$$\text{Integrating } lx + my + nz = c_2 \text{ --- (2).}$$

$$\therefore \text{Solution is } \underline{f(x^2 + y^2 + z^2, lx + my + nz) = 0}$$

$$\textcircled{2} \left( \frac{y-z}{yz} \right) p + \left( \frac{z-x}{zx} \right) q = \frac{x-y}{xy}$$

$$P = \frac{y-z}{yz}, \quad Q = \frac{z-x}{zx}, \quad R = \frac{x-y}{xy}$$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad (\text{S.E})$$

$$\frac{yz dx}{y-z} = \frac{zx dy}{z-x} = \frac{xy dz}{x-y}$$

Choose the multiplier (1,1,1)

$$\frac{yz dx}{y-z} = \frac{zx dy}{z-x} = \frac{xy dz}{x-y} = \frac{yz dx + zx dy + xy dz}{0}$$

$$\text{ie } yz dx + zx dy + xy dz = 0$$

$$\text{ie } d(xyz) = 0 \implies \underline{xyz = c_1} \text{ --- (1)}$$

Choose the multiplier  $(x, y, z)$

$$\frac{yz \, dx}{y-z} = \frac{zx \, dy}{z-x} = \frac{xy \, dz}{x-y} = \frac{xyz \, dx + yz \, x \, dy + xy \, z \, dz}{x(y-z) + y(z-x) + z(x-y)}$$

$$= \frac{xyz (dx + dy + dz)}{0}$$

$$\therefore xyz (dx + dy + dz) = 0$$

$$xyz \neq 0 \quad dx + dy + dz = 0$$

$$\text{Integrating } x + y + z = C_2 \quad \text{--- (2)}$$

$\therefore$  Solution is  $f(xyz, x+y+z) = 0$ .

③  $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

$$P = x^2(y-z), \quad Q = y^2(z-x), \quad R = z^2(x-y)$$

$$\text{S.E} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

Using multipliers  $\frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{z^2}$  we get

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz}{y-z + z-x + x-y}$$

$$= \frac{\frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz}{0}$$

$$\Rightarrow \frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz = 0$$

Integrating  $-\frac{1}{x} + -\frac{1}{y} + -\frac{1}{z} = C_1$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -C_1 = a \text{ (say)} \quad \text{--- (1)}$$

Again Using multipliers  $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$

we get

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{0}$$

$$\left\{ \begin{aligned} &\frac{1}{x} (x^2(y-z)) + \frac{1}{y} (y^2(z-x)) + \frac{1}{z} (z^2(x-y)) \\ &= xy - xz + yz - yx + zx - zy \\ &= 0 \end{aligned} \right.$$

$$\therefore \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0.$$

Integrating  $\log x + \log y + \log z = \log b.$

$$\log (xyz) = \log b.$$

$$\therefore xyz = b \quad \text{--- (2)}$$

(1) and (2) are independent solutions of subsidiary equations.

$\therefore$  Solution of given p.d.e

$$\text{is } \underline{\underline{\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = f(xyz)}}$$

④ Solve  $(y+zx)P - (x+yz)Q = x^2 - y^2$

$$\text{S.E} \Rightarrow \frac{dx}{y+zx} = \frac{dy}{-(x+yz)} = \frac{dz}{x^2-y^2}$$

Using multipliers  $x, y, -z$ , we get

$$\begin{aligned} \frac{dx}{y+zx} &= \frac{dy}{-x-yz} = \frac{dz}{x^2-y^2} = \frac{x dx + y dy - z dz}{x(y+zx) + y(-x-yz) - z(x^2-y^2)} \\ &= \frac{x dx + y dy - z dz}{0} \end{aligned}$$

$$\therefore x dx + y dy - z dz = 0$$

$$\begin{aligned} \text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} - \frac{z^2}{2} &= a \\ \Rightarrow x^2 + y^2 - z^2 &= 2a = C_1 \quad \text{--- (1)} \end{aligned}$$

Using multipliers  $y, x, 1$

$$\begin{aligned} \frac{dx}{y+zx} &= \frac{dy}{-x-yz} = \frac{dz}{x^2-y^2} = \frac{y dx + x dy + dz}{y(y+zx) + x(-x-yz) + x^2 - y^2} \\ &= \frac{y dx + x dy + dz}{0} \end{aligned}$$

$$\Rightarrow y dx + x dy + dz = 0$$

$$d(xy) + dz = 0$$

$$\text{Integrating } xy + z = b \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(x^2 + y^2 - z^2, xy + z) = 0$$

⑤ Solve  $x(y-z)p + y(z-x)q = z(x-y)$

$$\text{S.E} \Rightarrow \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$$

Using multiplier  $(1, 1, 1)$  we get

$$\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)} = \frac{dx+dy+dz}{0}$$

$$\therefore dx+dy+dz=0$$

$$\text{Integrating } x+y+z = a \quad \text{--- (1)}$$

Using multipliers  $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$

$$\begin{aligned} \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)} &= \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{y-z + z-x + x-y} \\ &= \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0} \end{aligned}$$

$$\therefore \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz = 0$$

$$\log x + \log y + \log z = \log b.$$

$$\log(xyz) = \log b$$

$$\therefore xyz = b. \quad \text{--- (2)}$$

① and ② are independent solutions of S.E

$\therefore$  Solution is  $f(x+y+z, xyz) = 0$

⑥ Solve  $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$

$P = z^2 - 2yz - y^2$ ,  $Q = xy + zx$ ,  $R = xy - zx$

S.E  $\Rightarrow \frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx}$

Take the last two fractions

$$\frac{dy}{xy + zx} = \frac{dz}{xy - zx} \quad \{\text{Cancel } x\}$$

$$\frac{dy}{y+z} = \frac{dz}{y-z}$$

$$(y-z)dy = (y+z)dz$$

$$y dy - z dz = y dz + z dy$$

Rearranging  $\Rightarrow y dy - z dz = y dz + z dy$

$$y dy - z dz = d(zy)$$

Integrating

$$\frac{y^2}{2} - \frac{z^2}{2} = zy + a.$$

$$\frac{y^2 - z^2}{2} - zy = a. \quad \text{--- (1)}$$

To find the second independent solution

choose the multiplier  $(x, y, z)$

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx} = \frac{x dx + y dy + z dz}{0}$$

$$\Rightarrow x dx + y dy + z dz = 0$$

$$\text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = b.$$

$$x^2 + y^2 + z^2 = 2b = c_1 \text{ — (2).}$$

$$\therefore \text{Solution is } f\left(\frac{y^2 - z^2}{2} - zy, x^2 + y^2 + z^2\right) = 0$$

$$(7) \text{ Solve } pz - qz = z^2 + (x+y)^2$$

$$P = z, \quad Q = -z, \quad R = z^2 + (x+y)^2$$

$$\text{S.E.} \Rightarrow \frac{dx}{z} = \frac{dy}{-z} = \frac{dz}{z^2 + (x+y)^2}$$

From first two fractions

$$\frac{dx}{z} = \frac{dy}{-z} \Rightarrow dx = -dy$$

$$\text{Integrating } x = -y + c_1$$

$$\Rightarrow x + y = c_1 \text{ — (1).}$$

Take the first and last fractions, we

$$\text{get } \frac{dx}{z} = \frac{dz}{z^2 + (x+y)^2}$$

$$\text{Using (1) } x + y = c_1$$

$$\frac{dx}{z} = \frac{dz}{z^2 + c_1^2} \Rightarrow dx = \frac{z dz}{z^2 + c_1^2}$$

$$\therefore dx = \frac{1}{2} \left[ \frac{2z dz}{z^2 + c_1^2} \right]$$

Integrating

$$x = \frac{1}{2} \log(z^2 + c_1^2) + c_2$$

$$= \frac{1}{2} \log(z^2 + (x+y)^2) + c_2 \quad \left\{ \begin{array}{l} \text{put } c_1 = x+y \end{array} \right.$$

$$x - \frac{1}{2} \log(z^2 + (x+y)^2) = c_2 \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(x+y, x - \frac{1}{2} \log(z^2 + (x+y)^2)) = 0.$$

⑧ Solve  $xp - yq = y^2 - x^2$

$$P = x, \quad Q = -y, \quad R = y^2 - x^2$$

$$S.E \Rightarrow \frac{dx}{x} = \frac{dy}{-y} = \frac{dz}{y^2 - x^2}$$

From first two we get

$$\frac{dx}{x} = \frac{dy}{-y}$$

$$\log x = -\log y + \log a$$

$$\log x + \log y = \log a$$

$$\log(xy) = \log a$$

$$xy = a \quad \text{--- (1)}$$

Use the multiplier  $(x, y, 1)$

$$\frac{dx}{x} = \frac{dy}{-y} = \frac{dz}{y^2 - x^2} = \frac{x dx + y dy + dz}{x^2 - y^2 + y^2 - x^2}$$

$$= \frac{x dx + y dy + dz}{0}$$

$$\Rightarrow x dx + y dy + dz = 0$$

$$\text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} + z = b$$

$$x^2 + y^2 + 2z = 2b = c \quad \text{--- (2)}$$

$$\therefore \text{Solution is } \underline{f(xy, x^2 + y^2 + 2z) = 0}$$

⑨ Solve  $P - 2Q = 3x^2 \sin(y + 2x)$   
 $P = 1, Q = -2, R = 3x^2 \sin(y + 2x)$

$$\frac{dx}{1} = \frac{dy}{-2} = \frac{dz}{3x^2 \sin(y + 2x)}$$

$$\text{From first two } \frac{dx}{1} = \frac{dy}{-2}$$

$$-2dx = dy$$

$$\text{Integrating } -2x = y + a$$

$$y + 2x = a \quad \text{--- (1)}$$

$$\text{Again consider } \frac{dx}{1} = \frac{dz}{3x^2 \sin(y + 2x)}$$

$$\frac{dx}{1} = \frac{dz}{3x^2 \sin a} \quad \{\text{Using (1)}\}$$

$$dz = 3x^2 \sin a \, dx$$

$$\text{Integrating } z = 3 \sin a \left[ \frac{x^3}{3} \right] + b$$

$$z = \sin a \cdot x^3 + b.$$

$$z = \sin(y+2x) x^3 + b \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(y+2x, z - \sin(y+2x) x^3) = 0$$

$$\textcircled{10} \text{ Solve } (y+z) \frac{\partial z}{\partial x} + (z+x) \frac{\partial z}{\partial y} = x+y$$

$$P = y+z, \quad Q = z+x, \quad R = x+y$$

$$\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} \quad \text{--- (1)}$$

To solve this, Consider the multipliers  $(1, -1, 0)$  and  $(0, 1, -1)$ .

$$\therefore \frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} = \frac{dx-dy}{y-x} = \frac{dy-dz}{z-y}$$

Take the last two fractions

$$\frac{dx-dy}{y-x} = \frac{dy-dz}{z-y}$$

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} \Rightarrow \frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z}$$

$$\text{Integrating we get } \log(x-y) = \log(y-z) + \log C_1$$

$$\text{ie } \log(x-y) - \log(y-z) = \log C_1$$

$$\log \left( \frac{x-y}{y-z} \right) = \log C_1 \quad (1)$$

$$\therefore \frac{x-y}{y-z} = C_1 \quad (2)$$

Consider the multipliers  $(1, 1, 1)$  and  $(1, -1, 0)$

$$\begin{aligned} (1) \Rightarrow \frac{dx}{y+z} &= \frac{dy}{z+x} = \frac{dz}{x+y} = \frac{dx+dy+dz}{2x+2y+2z} \\ &= \frac{dx-dy}{y-x} \end{aligned}$$

$$\therefore \frac{dx-dy}{y-x} = \frac{dx+dy+dz}{2x+2y+2z}$$

$$\frac{d(x-y)}{-(x-y)} = \frac{dx+dy+dz}{2(x+y+z)}$$

$$-2 \frac{d(x-y)}{x-y} = \frac{d(x+y+z)}{x+y+z}$$

$$\text{Integrating } -2 \log(x-y) = \log(x+y+z) + \log C_2$$

$$\begin{aligned} \log(x+y+z) + 2 \log(x-y) &= -\log C_2 \\ \log \{ (x+y+z)(x-y)^2 \} &= \log b \end{aligned}$$

$$\therefore (x+y+z)(x-y)^2 = b \quad (3)$$

(2) and (3) are independent solutions of S.F

$$\therefore \text{Solution is } f\left(\frac{x-y}{y-z}, (x+y+z)(x-y)^2\right) = 0$$

### H.W

$$\textcircled{1} y^2 p - xy q = x(z - 2y)$$

$$\textcircled{2} (xz + yz)p + (xz - yz)q = x^2 + y^2$$

$$\textcircled{3} (y^2 + z^2)p - xy q + xz = 0$$

$$\textcircled{4} \tan x p + \tan y q = \tan z$$

$$\textcircled{5} \sqrt{x} p + \sqrt{y} q = \sqrt{z}$$

Ktunotes.in

Lagrange's linear equations

A linear partial differential equation of the first order, commonly known as Lagrange's linear equations is of the form

$$P \frac{\partial z}{\partial x} + Q \frac{\partial z}{\partial y} = R$$

ie  $Pp + Qq = R$  <sup>①</sup>, where  $P, Q$  and  $R$  are functions of  $x, y$  and  $z$ .

To solve the equation ①, Consider the subsidiary equations,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \text{ --- ②}$$

$U(x, y, z) = a$  and  $V(x, y, z) = b$  are any two independent solutions of the above subsidiary equation ②

Then the complete solution of ① is

$$f(u, v) = 0 \quad \text{or} \quad u = f(v) \\ \text{or} \quad v = f(u).$$

Following methods are used for solving  
Subsidiary equations.

Method-1 : Method of grouping.

$$S.E \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

we can eliminate  $z$  from first two fractions of S.E, then a solution  $u(x, y) = a$  can be obtained by direct integration. Apply the same procedure for the other two pairs of fractions we get the second independent solution.

eg:  $\triangleright$  Solve  $y^2 z : \frac{\partial z}{\partial x} + x^2 z \frac{\partial z}{\partial y} = xy^2$

$$\text{ie } y^2 z p + x^2 z q = R.$$

$$P = y^2 z, \quad Q = x^2 z, \quad R = xy^2$$

Form the subsidiary equation.

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

$$\frac{dx}{y^2 z} = \frac{dy}{x^2 z} = \frac{dz}{xy^2} \quad (S.E)$$

Grouping first two fractions.

$$\text{ie } \frac{dx}{y^2 z} = \frac{dy}{x^2 z}$$

$$\Rightarrow \frac{dx}{y^2} = \frac{dy}{x^2}$$

$$\text{ie } x^2 dx = y^2 dy$$

Integrate both sides.

$$\int x^2 dx = \int y^2 dy$$

$$\frac{x^3}{3} = \frac{y^3}{3} + C_1$$

$$x^3 - y^3 = 3C_1 \\ = a \text{ (say)}$$

$$x^3 - y^3 = a \quad \text{--- (1)} \quad \{u=a\}.$$

Grouping first and last fractions in S.E

$$\frac{dx}{y^2 z} = \frac{dz}{xy^2}$$

$$\Rightarrow \frac{dx}{z} = \frac{dz}{x}$$

$$x dx = z dz$$

$$\text{Integrating } \frac{x^2}{2} = \frac{z^2}{2} + C_2$$

$$x^2 - z^2 = 2C_2 = b \quad \text{--- (2)} \quad \{v=b\}$$

(1) and (2) are two independent solutions of subsidiary equation.

∴ Solution of the given Lagrange's equation

is  $f(u, v) = 0$

ie  $f(x^3 - y^3, x^2 - z^2) = 0$

or  $x^3 - y^3 = f(x^2 - z^2)$

or  $x^2 - z^2 = f(x^3 - y^3)$

② Solve  $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = z^2$

ie  $x^2 p + y^2 q = z^2$

$P = x^2$     $Q = y^2$    ,    $R = z^2$

S.E  $\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{z^2}$$

Grouping first and second fractions

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

Integrating  $\int \frac{dx}{x^2} = \int \frac{dy}{y^2}$

$$-\frac{1}{x} = -\frac{1}{y} + a.$$

$$\frac{1}{y} - \frac{1}{x} = a \quad \text{--- (1)}$$

Grouping last two fractions,

$$\frac{dy}{y^2} = \frac{dz}{z^2}$$

Integrating  $-\frac{1}{y} = -\frac{1}{z} + b.$

ie  $\frac{1}{z} - \frac{1}{y} = b. \quad \text{--- (2)}$

① and ② are ~~linearly~~ independent solutions of S.E

$\therefore$  Solution is  $f\left(-\frac{1}{x} + \frac{1}{y}, \frac{1}{z} - \frac{1}{y}\right) = 0$

or  $-\frac{1}{x} + \frac{1}{y} = f\left(\frac{1}{z} - \frac{1}{y}\right)$

or  $\frac{1}{z} - \frac{1}{y} = f\left(-\frac{1}{x} + \frac{1}{y}\right)$

③ Solve  $zp = -x. \quad (Pp + Qq = R)$

$P = z, Q = 0, R = -x.$

$$\frac{dx}{p} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{z} = \frac{dy}{0} = \frac{dz}{-x}$$

Grouping first two fractions,

$$\frac{dx}{z} = \frac{dy}{0} \Rightarrow z dy = 0$$

$z \neq 0 \Rightarrow dy = 0$

Grouping first and last

$$\frac{dx}{z} = \frac{dz}{-x} \Rightarrow x dz = -x dx$$

$y = C_1$  (Const)  
--- ①.

Integrating  $\frac{z^2}{2} = -\frac{x^2}{2} + C_2$

$$\frac{x^2}{2} + \frac{z^2}{2} = C_2$$

$$x^2 + z^2 = 2C_2 = b \cdot (\sec y) \quad \text{--- (2)}$$

$\therefore$  Solution is  $f(y, x^2 + z^2) = 0$ .

(4)  $P - Q = \log(x+y)$

$$P = 1, \quad Q = -1, \quad R = \log(x+y)$$

$$\text{S.E} \Rightarrow \frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{\log(x+y)}$$

Comparing first two fractions,

$$\frac{dx}{1} = \frac{dy}{-1} \Rightarrow -dx = dy$$

$$\text{Integrating} \Rightarrow -x = y + a$$

$$x + y = a \quad \text{--- (1)}$$

$$\frac{dx}{1} = \frac{dz}{\log(x+y)}$$

$$\Rightarrow \frac{dx}{1} = \frac{dz}{\log a} \quad (\text{using (1)})$$

$$dx = \frac{1}{\log a} dz$$

$$\text{Integrating} \quad x = \frac{z}{\log a} + b$$

$$\text{ie } x = \frac{z}{\log(x+y)} + b.$$

$$\text{ie } x - \frac{z}{\log(x+y)} = b. \text{---(2)}$$

① and ② are independent solution of subsidiary Equations.

$$\therefore \text{Solution is } \underline{f\left(x+y, x - \frac{z}{\log(x+y)}\right) = 0}$$

$$\textcircled{5} \quad xp + yq = 3z. \quad \{Pp + Qq = R\}.$$

$$P = x, \quad Q = y, \quad R = 3z.$$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{3z}$$

From first two fractions,

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating we get  $\log x = \log y + \log a$   
(Take the constant also  $\log a$ )

$$\therefore \log x - \log y = \log a.$$

$$\log\left(\frac{x}{y}\right) = \log a.$$

$$\text{ie } \frac{x}{y} = a \text{ --- (1). } \{u=a\}$$

$$\text{Take last two fractions, } \frac{dy}{y} = \frac{dz}{3z}$$

Integrating  $\log y = \frac{1}{3} \log z + \log b.$

$$\log y - \frac{1}{3} \log z = \log b.$$

$$\log \left\{ \frac{y}{z^{1/3}} \right\} = \log b.$$

$$\text{ie } \frac{y}{z^{1/3}} = b. \text{ --- (2) } \{v=b\}$$

$\therefore$  (1) and (2) are two independent solutions of subsidiary equation.

$\therefore$  solution is

$$\underline{\underline{\frac{x}{y} = f\left(\frac{y}{z^{1/3}}\right)}}$$

$$\left\{ \begin{array}{l} \text{Solution} \\ f(u, v) = 0 \\ \text{or} \\ u = f(v) \\ \text{or } v = f(u) \end{array} \right\}$$

(6)  $a(p+q) = z \implies ap + aq = z$

$$P=a, Q=a, R=z.$$

$$\frac{dx}{a} = \frac{dy}{a} = \frac{dz}{z}$$

$$\frac{dx}{a} = \frac{dy}{a} \implies dx = dy$$

$$\implies x = y + a$$

$$x - y = a \text{ --- (1)}$$

$$\frac{dy}{a} = \frac{dz}{z}$$

$$\text{Integrating } \frac{1}{a} y = \log z + b.$$

$$\frac{y}{a} - \log z = b \quad \text{--- (2)}$$

(5)

$\therefore$  Solution is

$$f(x-y, \frac{y}{a} - \log z) = 0$$

Method 2 : Method of Multipliers.

Choose any three multipliers  $l, m, n$  (which may be constant or functions of  $x, y, z$ )

Such that in

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{lP + mQ + nR},$$

the expression  $lP + mQ + nR = 0$ ,

then we get  $l dx + m dy + n dz = 0$ .

Integrating we get first solution  $u(x, y, z) = a$

Another independent solution  $v(x, y, z) = b$  can be found by choosing another set of multipliers  $l', m', n'$  such that  $l'P + m'Q + n'R = 0$ .

① Solve  $(mz - ny) \frac{\partial z}{\partial x} + (nx - lz) \frac{\partial z}{\partial y} = ly - mx$

$$P = mz - ny, \quad Q = nx - lz, \quad R = ly - mx$$

$$S.E \Rightarrow \frac{dx}{xz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx}$$

Choose the multiplier  $x, y, z$ .

$$\text{ie } \frac{dx}{xz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{x dx + y dy + z dz}{0}$$

$$\begin{cases} x(xz-ny) + y(nx-lz) + z(ly-mx) \\ = x^2z - xny + ynx - ylz + zly - mxz \\ = 0 \end{cases}$$

Take first and fourth fraction

$$\frac{dx}{xz-ny} = \frac{x dx + y dy + z dz}{0}$$

$$\text{ie } (xz-ny)(x dx + y dy + z dz) = 0$$

$$xz-ny \neq 0 \Rightarrow x dx + y dy + z dz = 0$$

$$\text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = a$$

$$\underline{x^2 + y^2 + z^2 = 2a = c_1} \quad \text{--- (1)}$$

② To find the second independent solution choose the multipliers  $l, m, n$ .

$$\text{ie } \frac{dx}{xz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{l dx + m dy + n dz}{l(xz-ny) + m(nx-lz) + n(ly-mx)} \Rightarrow 0$$

Take the first and last fractions

⑥

$$\frac{dx}{xz-ny} = \frac{l dx + m dy + n dz}{0}$$

$$\text{ie } (xz-ny)(l dx + m dy + n dz) = 0$$

$$\text{ie } l dx + m dy + n dz = 0$$

$$\text{Integrating } lx + my + nz = c_2 \text{ --- (2).}$$

$$\therefore \text{Solution is } \underline{f(x^2 + y^2 + z^2, lx + my + nz) = 0}$$

$$\textcircled{2} \left( \frac{y-z}{yz} \right) P + \left( \frac{z-x}{zx} \right) Q = \frac{x-y}{xy}$$

$$P = \frac{y-z}{yz}, \quad Q = \frac{z-x}{zx}, \quad R = \frac{x-y}{xy}$$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad (\text{S.E})$$

$$\frac{yz dx}{y-z} = \frac{zx dy}{z-x} = \frac{xy dz}{x-y}$$

Choose the multiplier (1,1,1)

$$\frac{yz dx}{y-z} = \frac{zx dy}{z-x} = \frac{xy dz}{x-y} = \frac{yz dx + zx dy + xy dz}{0}$$

$$\text{ie } yz dx + zx dy + xy dz = 0$$

$$\text{ie } d(xyz) = 0 \implies \underline{xyz = c_1} \text{ --- (1)}$$

Choose the multiplier  $(x, y, z)$

$$\frac{yz \, dx}{y-z} = \frac{zx \, dy}{z-x} = \frac{xy \, dz}{x-y} = \frac{xyz \, dx + yz \, x \, dy + xy \, z \, dz}{x(y-z) + y(z-x) + z(x-y)}$$

$$= \frac{xyz (dx + dy + dz)}{0}$$

$$\therefore xyz (dx + dy + dz) = 0$$

$$xyz \neq 0 \quad dx + dy + dz = 0$$

$$\text{Integrating } x + y + z = C_2 \quad \text{--- (2)}$$

$$\therefore \text{Solution is } \underline{f(xyz, x+y+z) = 0}.$$

③  $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

$$P = x^2(y-z), \quad Q = y^2(z-x), \quad R = z^2(x-y)$$

$$\text{S.E} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

Using multipliers  $\frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{z^2}$  we get

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz}{y-z + z-x + x-y}$$

$$= \frac{\frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz}{0}$$

$$\Rightarrow \frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz = 0$$

Integrating  $-\frac{1}{x} + -\frac{1}{y} + -\frac{1}{z} = C_1$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -C_1 = a \text{ (say)} \quad \text{--- (1)}$$

Again Using multipliers  $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$

we get

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{0}$$

$$\left\{ \begin{aligned} &\frac{1}{x} (x^2(y-z)) + \frac{1}{y} (y^2(z-x)) + \frac{1}{z} (z^2(x-y)) \\ &= xy - xz + yz - yx + zx - zy \\ &= 0 \end{aligned} \right.$$

$$\therefore \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0.$$

Integrating  $\log x + \log y + \log z = \log b.$

$$\log (xyz) = \log b.$$

$$\therefore xyz = b \quad \text{--- (2)}$$

(1) and (2) are independent solutions of subsidiary equations.

$\therefore$  Solution of given p.d.e

$$\text{is } \underline{\underline{\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = f(xyz)}}$$

④ Solve  $(y+zx)P - (x+yz)Q = x^2 - y^2$

$$S.E \Rightarrow \frac{dx}{y+zx} = \frac{dy}{-(x+yz)} = \frac{dz}{x^2-y^2}$$

Using multipliers  $x, y, -z$ , we get

$$\begin{aligned} \frac{dx}{y+zx} &= \frac{dy}{-x-yz} = \frac{dz}{x^2-y^2} = \frac{x dx + y dy - z dz}{x(y+zx) + y(-x-yz) - z(x^2-y^2)} \\ &= \frac{x dx + y dy - z dz}{0} \end{aligned}$$

$$\therefore x dx + y dy - z dz = 0$$

$$\begin{aligned} \text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} - \frac{z^2}{2} &= a \\ \Rightarrow x^2 + y^2 - z^2 &= 2a = C_1 \quad \text{--- (1)} \end{aligned}$$

Using multipliers  $y, x, 1$

$$\begin{aligned} \frac{dx}{y+zx} &= \frac{dy}{-x-yz} = \frac{dz}{x^2-y^2} = \frac{y dx + x dy + dz}{y(y+zx) + x(-x-yz) + x^2 - y^2} \\ &= \frac{y dx + x dy + dz}{0} \end{aligned}$$

$$\Rightarrow y dx + x dy + dz = 0$$

$$d(xy) + dz = 0$$

$$\text{Integrating } xy + z = b \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(x^2 + y^2 - z^2, xy + z) = 0$$

⑤ Solve  $x(y-z)p + y(z-x)q = z(x-y)$

$$S.E \Rightarrow \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$$

Using multiplier  $(1, 1, 1)$  we get

$$\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)} = \frac{dx+dy+dz}{0}$$

$$\therefore dx+dy+dz=0$$

$$\text{Integrating } x+y+z = a \quad \text{--- (1)}$$

Using multipliers  $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$

$$\begin{aligned} \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)} &= \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{y-z+z-x+x-y} \\ &= \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0} \end{aligned}$$

$$\therefore \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz = 0$$

$$\log x + \log y + \log z = \log b.$$

$$\log(xyz) = \log b$$

$$\therefore xyz = b. \quad \text{--- (2)}$$

① and ② are independent solutions of S.E

$\therefore$  Solution is  $f(x+y+z, xyz) = 0$

⑥ Solve  $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$

$P = z^2 - 2yz - y^2$ ,  $Q = xy + zx$ ,  $R = xy - zx$

S.E  $\Rightarrow \frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx}$

Take the last two fractions

$\frac{dy}{xy + zx} = \frac{dz}{xy - zx}$  {Cancel  $x$ }

$\frac{dy}{y+z} = \frac{dz}{y-z}$

$(y-z)dy = (y+z)dz$

$y dy - z dz = y dz + z dy$

Rearranging  $\Rightarrow y dy - z dz = y dz + z dy$

$y dy - z dz = d(zy)$

Integrating

$\frac{y^2}{2} - \frac{z^2}{2} = zy + a.$

$\frac{y^2 - z^2}{2} - zy = a. \text{--- (1)}$

To find the second independent solution

choose the multiplier  $(x, y, z)$

$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx} = \frac{x dx + y dy + z dz}{0}$

$$\Rightarrow x dx + y dy + z dz = 0$$

$$\text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = b.$$

$$x^2 + y^2 + z^2 = 2b = c_1 \text{ — (2).}$$

$$\therefore \text{Solution is } f\left(\frac{y^2 - z^2}{2} - zy, x^2 + y^2 + z^2\right) = 0$$

$$(7) \text{ Solve } pz - qz = z^2 + (x+y)^2$$

$$P = z, Q = -z, R = z^2 + (x+y)^2$$

$$\text{S.E.} \Rightarrow \frac{dx}{z} = \frac{dy}{-z} = \frac{dz}{z^2 + (x+y)^2}$$

From first two fractions

$$\frac{dx}{z} = \frac{dy}{-z} \Rightarrow dx = -dy$$

$$\text{Integrating } x = -y + c_1$$

$$\Rightarrow x + y = c_1 \text{ — (1).}$$

Take the first and last fractions, we

$$\text{get } \frac{dx}{z} = \frac{dz}{z^2 + (x+y)^2}$$

$$\text{Using (1) } x + y = c_1$$

$$\frac{dx}{z} = \frac{dz}{z^2 + c_1^2} \Rightarrow dx = \frac{z dz}{z^2 + c_1^2}$$

$$\therefore dx = \frac{1}{2} \left[ \frac{2z dz}{z^2 + c_1^2} \right]$$

Integrating

$$x = \frac{1}{2} \log(z^2 + c_1^2) + c_2$$

$$= \frac{1}{2} \log(z^2 + (x+y)^2) + c_2 \quad \left\{ \begin{array}{l} \text{put } c_1 = x+y \end{array} \right.$$

$$x - \frac{1}{2} \log(z^2 + (x+y)^2) = c_2 \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(x+y, x - \frac{1}{2} \log(z^2 + (x+y)^2)) = 0.$$

⑧ Solve  $xp - yq = y^2 - x^2$

$$P = x, \quad Q = -y, \quad R = y^2 - x^2$$

$$S.E \Rightarrow \frac{dx}{x} = \frac{dy}{-y} = \frac{dz}{y^2 - x^2}$$

From first two we get

$$\frac{dx}{x} = \frac{dy}{-y}$$

$$\log x = -\log y + \log a$$

$$\log x + \log y = \log a$$

$$\log(xy) = \log a$$

$$xy = a \quad \text{--- (1)}$$

Use the multiplier  $(x, y, 1)$

$$\frac{dx}{x} = \frac{dy}{-y} = \frac{dz}{y^2 - x^2} = \frac{x dx + y dy + dz}{x^2 - y^2 + y^2 - x^2}$$

$$= \frac{x dx + y dy + dz}{0}$$

$$\Rightarrow x dx + y dy + dz = 0$$

$$\text{Integrating } \frac{x^2}{2} + \frac{y^2}{2} + z = b$$

$$x^2 + y^2 + 2z = 2b = c \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(xy, x^2 + y^2 + 2z) = 0$$

⑨ Solve  $P - 2Q = 3x^2 \sin(y + 2x)$

$$P = 1, \quad Q = -2, \quad R = 3x^2 \sin(y + 2x)$$

$$\frac{dx}{1} = \frac{dy}{-2} = \frac{dz}{3x^2 \sin(y + 2x)}$$

$$\text{From first two } \frac{dx}{1} = \frac{dy}{-2}$$

$$-2dx = dy$$

$$\text{Integrating } -2x = y + a$$

$$y + 2x = a \quad \text{--- (1)}$$

$$\text{Again consider } \frac{dx}{1} = \frac{dz}{3x^2 \sin(y + 2x)}$$

$$\frac{dx}{1} = \frac{dz}{3x^2 \sin a} \quad \{\text{Using (1)}\}$$

$$dz = 3x^2 \sin a \, dx$$

$$\text{Integrating } z = 3 \sin a \left[ \frac{x^3}{3} \right] + b$$

$$z = \sin a \cdot x^3 + b.$$

$$z = \sin(y+2x) x^3 + b \quad \text{--- (2)}$$

$$\therefore \text{Solution is } f(y+2x, z - \sin(y+2x) x^3) = 0$$

$$\textcircled{10} \text{ Solve } (y+z) \frac{\partial z}{\partial x} + (z+x) \frac{\partial z}{\partial y} = x+y$$

$$P = y+z, \quad Q = z+x, \quad R = x+y$$

$$\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} \quad \text{--- (1)}$$

To solve this, Consider the multipliers  $(1, -1, 0)$  and  $(0, 1, -1)$ .

$$\therefore \frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} = \frac{dx-dy}{y-x} = \frac{dy-dz}{z-y}$$

Take the last two fractions

$$\frac{dx-dy}{y-x} = \frac{dy-dz}{z-y}$$

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} \Rightarrow \frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z}$$

$$\text{Integrating we get } \log(x-y) = \log(y-z) + \log C_1$$

$$\text{i.e. } \log(x-y) - \log(y-z) = \log C_1$$

$$\log \left( \frac{x-y}{y-z} \right) = \log C_1 \quad (1)$$

$$\therefore \frac{x-y}{y-z} = C_1 \quad (2)$$

Consider the multipliers  $(1, 1, 1)$  and  $(1, -1, 0)$

$$\begin{aligned} (1) \Rightarrow \frac{dx}{y+z} &= \frac{dy}{z+x} = \frac{dz}{x+y} = \frac{dx+dy+dz}{2x+2y+2z} \\ &= \frac{dx-dy}{y-x} \end{aligned}$$

$$\therefore \frac{dx-dy}{y-x} = \frac{dx+dy+dz}{2x+2y+2z}$$

$$\frac{d(x-y)}{- (x-y)} = \frac{dx+dy+dz}{2(x+y+z)}$$

$$-2 \frac{d(x-y)}{x-y} = \frac{d(x+y+z)}{x+y+z}$$

$$\text{Integrating } -2 \log(x-y) = \log(x+y+z) + \log C_2$$

$$\begin{aligned} \log(x+y+z) + 2 \log(x-y) &= -\log C_2 \\ \log \{ (x+y+z)(x-y)^2 \} &= \log b \end{aligned}$$

$$\therefore (x+y+z)(x-y)^2 = b \quad (3)$$

(2) and (3) are independent solutions of S.F

$$\therefore \text{Solution is } \underline{\underline{f\left(\frac{x-y}{y-z}, (x+y+z)(x-y)^2\right) = 0}}$$

### H.W

$$\textcircled{1} y^2 p - xy q = x(z - 2y)$$

$$\textcircled{2} (xz + yz)p + (xz - yz)q = x^2 + y^2$$

$$\textcircled{3} (y^2 + z^2)p - xyq + xz = 0$$

$$\textcircled{4} \tan x p + \tan y q = \tan z$$

$$\textcircled{5} \sqrt{x} p + \sqrt{y} q = \sqrt{z}$$

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## Charpit's Method

It is a general method for solving non-linear first order partial differential equations,  $f(x, y, z, p, q) = 0$ .

### Working Rule

Step I :- Consider the non-linear p.d.e

$$f(x, y, z, p, q) = 0. \text{ --- (1)}$$

Step II :- Differentiate (1) partially

w.r.to  $x, y, z, p, q$ .

i.e we get  $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}, \frac{\partial f}{\partial p}, \frac{\partial f}{\partial q}$

Step III :- Find charpit's subsidiary equations or Auxiliary equations.

$$\frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}} = \frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}}$$

Step IV :-

Solving subsidiary equations and eqn (1)

find the values of  $p$  and  $q$  in terms

of  $x, y$  and  $z$ . i.e  $p = p(x, y, z)$

$$q = q(x, y, z)$$

Step V :-

Complete solution of (1) will be given by

$$dz = p dx + q dy.$$

Integrating both sides, we get the solution.

① Solve by Charpit's method  $p^2 - q^2 = x - y$

$$f(x, y, z, p, q) = 0 \quad \text{ie}$$

$$f(x, y, z, p, q) = p^2 - q^2 - x + y = 0 \quad \text{--- ①.}$$

Diff. partially ① w.r.t.  $x, y, z, p, q$

$$\frac{\partial f}{\partial x} = -1, \quad \frac{\partial f}{\partial y} = 1.$$

$$\frac{\partial f}{\partial z} = 0, \quad \frac{\partial f}{\partial p} = 2p, \quad \frac{\partial f}{\partial q} = -2q$$

S.E  $\Rightarrow$

$$\begin{aligned} \frac{dx}{-\frac{\partial f}{\partial p}} &= \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}} = \frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}} \\ &= \frac{dq}{\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}} \end{aligned}$$

$\Rightarrow$

$$\frac{dx}{-2p} = \frac{dy}{2q} = \frac{dz}{-2p^2 + 2q^2} = \frac{dp}{-1} = \frac{dq}{1} \quad \text{--- ②}$$

Solving subsidiary equations.

Take 1<sup>st</sup> and 4<sup>th</sup>

$$\frac{dx}{-2p} = \frac{dp}{-1} \Rightarrow -dx = -2p dp$$

$$dx = 2p dp$$

Integrating

$$\int dx = \int 2p dp$$

$$x = p^2 + a \Rightarrow p^2 = x - a$$

$$p = \pm \sqrt{x-a}$$

Take the 2<sup>nd</sup> and last fractions from S.E

we get  $\frac{dy}{2q} = \frac{dq}{1}$

$$dy = 2q \cdot dq$$

$$\int dy = \int 2q dq$$

$$y = \frac{2q^2}{2} + b$$

$$q^2 = y - b \Rightarrow q = \pm \sqrt{y-b}$$

Substitute  $p$  and  $q$  in  $dz = p dx + q dy$

$$dz = \pm \sqrt{x-a} dx \pm \sqrt{y-b} dy$$

Integrating  $\int dz = \pm \int \sqrt{x-a} dx \pm \int \sqrt{y-b} dy$

$$z = \pm \frac{(x-a)^{3/2}}{3/2} \pm \frac{(y-b)^{3/2}}{3/2}$$

$$= \pm \frac{2}{3} (x-a)^{3/2} \pm \frac{2}{3} (y-b)^{3/2}$$

which is the required complete solution of the given p.d.e

$$(2) \text{ Solve } (p^2 + q^2)y = z^2.$$

$$f(x, y, z, p, q) = 0$$

$$\Rightarrow f(x, y, z, p, q) = (p^2 + q^2)y - z^2 = 0$$

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = p^2 + q^2$$

$$\frac{\partial f}{\partial z} = -2z, \quad \frac{\partial f}{\partial p} = 2py$$

$$\frac{\partial f}{\partial q} = 2qy - z.$$

$$\text{S.E} \Rightarrow$$

$$\frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-\frac{\partial f}{\partial z}} = \frac{dp}{\frac{\partial f}{\partial p} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial q} + q \frac{\partial f}{\partial z}}$$

$$\frac{dx}{-2py} = \frac{dy}{-(2qy - z)} = \frac{dz}{-p(2py) - q(2qy - z)}$$

$$= \frac{dp}{-pq} = \frac{dq}{p^2 + q^2 - q^2}$$

$$\Rightarrow \frac{dx}{-2py} = \frac{dy}{z - 2qy} = \frac{dz}{-2p^2y - 2q^2y - z^2} = \frac{dp}{-pq} = \frac{dq}{p^2} \quad \text{--- (1)}$$

$$\text{Take last two we get } \frac{dp}{-pq} = \frac{dq}{p^2}$$

Cancel p on both sides

$$\frac{dp}{-q} = \frac{dq}{p} \Rightarrow p dp = -q dq$$

$$\text{Integrating } \frac{p^2}{2} = -\frac{q^2}{2} + a$$

$$p^2 + q^2 = 2a$$

$$p^2 = 2a - q^2$$

$$p = \pm \sqrt{2a - q^2}$$

Substituting  $p^2 + q^2 = 2a$  in the given eqn.

$$(p^2 + q^2)y - qz = 0$$

$$\text{we get } 2ay - qz = 0$$

$$q = \frac{2ay}{z}$$

$$\therefore p = \pm \sqrt{2a - q^2}$$

$$= \pm \sqrt{2a - \frac{4a^2y^2}{z^2}}$$

$$= \pm \frac{\sqrt{2az^2 - 4a^2y^2}}{z}$$

$$\therefore dz = p dx + q dy$$

$$dz = \pm \frac{\sqrt{2az^2 - 4a^2y^2}}{z} dx + \frac{2ay}{z} dy$$

$$z dz = \pm \sqrt{2az^2 - 4a^2y^2} dx + 2ay dy$$

$$z dz - 2ay dy = \pm \sqrt{2az^2 - 4a^2y^2} dx$$

$$\frac{z dz - 2ay dy}{\sqrt{2az^2 - 4a^2y^2}} = \pm dx \quad \text{--- (2)}$$

$$\text{put } 2az^2 - 4a^2y^2 = t^2$$

$$\text{differentiating } 4az dx - 8a^2y dy = 2t dt$$

$$4a(z dz - 2ay dy) = 2t dt$$

$$z dz - 2ay dy = \frac{2t}{4a} dt$$

$$= \frac{t}{2a} dt$$

$$\text{Eq. (2)} \Rightarrow$$

$$\frac{\frac{t}{2a} dt}{\sqrt{t^2}} = \pm dx$$

$$\frac{1}{2a} \frac{t}{t} dt = \pm dx$$

$$\frac{1}{2a} dt = \pm dx$$

$$\text{Integrating } \frac{1}{2a} t = \pm x + b$$

$$\text{ie } \frac{1}{2a} \sqrt{2az^2 - 4a^2y^2} = \pm x + b$$

$$\sqrt{2az^2 - 4a^2y^2} = \pm 2a(x + b)$$

$$= \pm 2ax + k$$

Squaring both sides

$$2az^2 - 4a^2y^2 = (2ax+k)^2$$

$$2az^2 = 4a^2y^2 + (2ax+k)^2$$

$$\underline{\underline{z^2 = \frac{1}{2a} [4a^2y^2 + (2ax+k)^2]}}$$

③ Solve  $px + qy = pz$  by Charpit's method.

$$f(x, y, z, p, q) = px + qy - pz = 0 \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial x} = p$$

$$\frac{\partial f}{\partial p} = x - z$$

$$\frac{\partial f}{\partial y} = q$$

$$\frac{\partial f}{\partial q} = y - z$$

$$\frac{\partial f}{\partial z} = 0$$

S.E  $\Rightarrow$

$$\frac{dx}{q-x} = \frac{dy}{p-y} = \frac{dz}{pq - px + qp - qy} = \frac{dp}{p} = \frac{dq}{q}$$

Take the last two

$$\frac{dp}{p} = \frac{dq}{q}$$

Integrating  $\log p = \log q + \log a$

$$\log\left(\frac{p}{q}\right) = \log a \Rightarrow \frac{p}{q} = a$$

$$p = aq$$

Substitute  $p = aq$  in ①

$$\Rightarrow aqx + qy = aq \cdot q$$

~~again~~ Cancelling  $q$  we get

$$ax + y = aq$$

$$\therefore q = \frac{ax + y}{a} = \underline{\underline{x + \frac{y}{a}}}$$

$$\text{also } p = ax + y$$

$$\therefore dz = p dx + q dy$$

$$\Rightarrow dz = (ax + y) dx + \left(x + \frac{y}{a}\right) dy$$

$$= ax dx + y dx + x dy + \frac{y}{a} dy$$

$$dz = ax dx + \frac{y}{a} dy + d(xy)$$

$$\text{Integrating } z = a \frac{x^2}{2} + \frac{1}{a} \frac{y^2}{2} + xy + b.$$

$$= \underline{\underline{\frac{x^2 a}{2} + \frac{y^2}{2a} + xy + b}}$$

④ Solve  $pxy + pq + qy - yz = 0$  by using Charpit's method.

$$f(x, y, z, p, q) = pxy + pq + qy - yz = 0 \quad \text{--- ①}$$

First find the partial derivatives of  $f$  w.r. to  $x, y, z, p$  and  $q$ .

$$\frac{\partial f}{\partial x} = py$$

$$\frac{\partial f}{\partial y} = px + q$$

$$\frac{\partial f}{\partial z} = -y$$

$$\frac{\partial f}{\partial p} = xy + q$$

$$\frac{\partial f}{\partial q} = p + y$$

S.E  $\Rightarrow$

$$\frac{dx}{-xy - q} = \frac{dy}{-p - y} = \frac{dz}{-p(xy + q) - q(p + y)}$$

$$= \frac{dp}{py - py} = \frac{dq}{px + q - qy}$$

Take the last two, we get

$$\frac{dp}{0} = \frac{dq}{px + q - qy}$$

$$\Rightarrow (px + q - qy) dp = 0$$

$$[\because px + q - qy \neq 0]$$

$$\Rightarrow dp = 0$$

$$\text{i.e. } \underline{\underline{p = a}}$$

Substitute  $p = a$  in ①

$$axy + aq + qy - yz = 0$$

$$q(a + y) = yz - axy$$

$$\therefore q = \frac{yz - axy}{a+y}$$

$$dz = p dx + q dy$$

$$dz = a dx + \left( \frac{yz - axy}{a+y} \right) dy$$

$$\begin{aligned} dz - a dx &= \left( \frac{yz - axy}{a+y} \right) dy \\ &= \left[ \frac{y(z - ax)}{a+y} \right] dy \end{aligned}$$

$$\frac{dz - a dx}{z - ax} = \left( \frac{y}{a+y} \right) dy$$

$$\frac{d(z - ax)}{z - ax} = \left( \frac{y}{a+y} \right) dy$$

Integrating both sides.

$$\int \frac{d(z - ax)}{z - ax} = \int \left( \frac{y}{a+y} \right) dy$$

$$\log(z - ax) = \int \left( \frac{y + a - a}{a+y} \right) dy$$

$$= \int \left( 1 - \frac{a}{a+y} \right) dy$$

$$= \int dy - \int \left( \frac{a}{a+y} \right) dy$$

$$\log(z - ax) = y - a \log(a+y) + \log b.$$

$$\log(z - ax) + a \log(a+y) - \log b = y$$

$$\log(z - ax) + \log(a+y)^a - \log b = y$$

$$\log \left\{ \frac{(z - ax)(a+y)^a}{b} \right\} = y$$

$$\therefore \frac{(z - ax)(a+y)^a}{b} = e^y$$

$$\therefore z - ax = \frac{b e^y}{(a+y)^a}$$

$$\therefore z = ax + \frac{b e^y}{(a+y)^a}$$

⑤ Solve  $pq = xy$  using Charpit's method.

$$f(x, y, z, p, q) = pq - xy = 0 \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial x} = -y$$

$$\frac{\partial f}{\partial y} = -x$$

$$\frac{\partial f}{\partial z} = 0$$

$$\frac{\partial f}{\partial p} = q$$

$$\frac{\partial f}{\partial q} = p$$

$$S.E \Rightarrow$$

$$\frac{dx}{-q} = \frac{dy}{-p} = \frac{dz}{-pq-qp} = \frac{dp}{-y} = \frac{dq}{-x}.$$

Take first and last

$$\frac{dx}{-q} = \frac{dq}{-x} \Rightarrow q dq = x dx$$

Integrating

$$\frac{q^2}{2} = \frac{x^2}{2} + a.$$

$$\frac{q^2}{2} - \frac{x^2}{2} = a$$

$$q^2 - x^2 = 2a$$

$$q^2 = x^2 + 2a.$$

$$q = \pm \sqrt{x^2 + 2a}$$

$$\textcircled{1} \Rightarrow pq = xy$$

$$\therefore p = \frac{xy}{q} = \frac{xy}{\pm \sqrt{x^2 + 2a}}$$

$$dz = p dx + q dy = \frac{xy}{\pm \sqrt{x^2 + 2a}} dx + \pm \sqrt{x^2 + 2a} dy$$

$$= \pm d(y \sqrt{x^2 + 2a})$$

Integrating

$$\int dz = \pm \int d(y \sqrt{x^2 + 2a})$$

$$\underline{\underline{z = \pm y \sqrt{x^2 + 2a} + b}}$$

⑥ Solve  $z^2 = pqxy$  using charpit's method

$$f(x, y, z, p, q) = z^2 - pqxy = 0$$

$$\begin{array}{l|l} \frac{\partial f}{\partial x} = -pqy & \frac{\partial f}{\partial p} = -qxy \\ \frac{\partial f}{\partial y} = -pqx & \frac{\partial f}{\partial q} = -pxy \\ \frac{\partial f}{\partial z} = 2z & \end{array}$$

S.E  $\Rightarrow$

$$\frac{dx}{qxy} = \frac{dy}{pxy} = \frac{dz}{pqxy + 2pxy} = \frac{dp}{-pqy + 2pz}$$

$$= \frac{dq}{-pqx + 2qz}$$

$$\left(\frac{1}{x}, -\frac{1}{y}\right) \Rightarrow \frac{dx}{qxy} = \frac{dy}{pxy} = \frac{\frac{1}{x} dx - \frac{1}{y} dy}{\frac{1}{x}(qxy) - \frac{1}{y}(pxy)}$$

$$= \frac{\frac{1}{x} dx - \frac{1}{y} dy}{qy - px} \quad \text{--- (1)}$$

$$\left(\frac{1}{p}, -\frac{1}{q}\right) \Rightarrow \frac{dp}{-pqy + 2pz} = \frac{dq}{-pqx + 2qz} = \frac{\frac{1}{p} dp - \frac{1}{q} dq}{-qy + 2z + px - 2z}$$

$$= \frac{\frac{1}{p} dp - \frac{1}{q} dq}{px - qy} \quad \text{--- (2)}$$

From (1) and (2)

$$\frac{\frac{1}{x} dx - \frac{1}{y} dy}{qy - px} = \frac{\frac{1}{p} dp - \frac{1}{q} dq}{px - qy}$$

Cancel  $\frac{1}{qy-px}$  on both sides.

$$\frac{1}{x} dx - \frac{1}{y} dy = \frac{1}{q} dq - \frac{1}{p} dp$$

Integrating

$$\log x - \log y = \log q - \log p + \log a^2$$

$$\log\left(\frac{x}{y}\right) = \log\left(\frac{q}{p}\right) + \log a^2$$

$$\log\left(\frac{x}{y}\right) - \log\left(\frac{q}{p}\right) = \log a^2$$

$$\log\left(\frac{x}{y} \times \frac{p}{q}\right) = \log a^2$$

$$\frac{px}{yq} = a^2$$

$$\therefore p = \frac{a^2 y q}{x}$$

Substitute  $p$  in  $z^2 = pq^x y$

$$z^2 = \frac{a^2 y q}{x} \times q \times x y = a^2 q^2 y^2$$

$$z = \pm a q y \Rightarrow z = a q y$$

$$q = \frac{z}{ay}$$

$$dz = p dx + q dy$$

$$dz = \frac{a^2 y q}{x} dx + \frac{z}{ay} dy$$

$$= \frac{a^2 y \times \frac{z}{ay}}{x} dx + \frac{z}{ay} dy \quad (\because q = \frac{z}{ay})$$

$$= \frac{az}{x} dx + \frac{z}{ay} dy$$

$$\frac{dz}{z} = a \frac{dx}{x} + \frac{1}{a} \frac{dy}{y}$$

Integrating

$$\log z = a \log x + \frac{1}{a} \log y + \log b$$

$$= \log x^a + \log (y)^{1/a} + \log b$$

$$= \log (x^a y^{1/a} b)$$

$$\therefore \underline{\underline{z = x^a y^{1/a} b}}$$

H.W

- ① Solve  $1 + p^2 = qz$ .
  - ② Solve  $2z + p^2 + qy + 2y^2 = 0$
  - ③ Solve  $q + xp = p^2$
- } by Charpit's method.

## Solution of PDE by the method of Separation of Variables.

Partial differential equations together with the boundary conditions constitute boundary value problems. Method of separation of variables is a powerful tool to solve such boundary value problems.

If the given pde contains 2 independent variables say  $x$  and  $y$  and one dependent variable  $u$ . In this method we assume

that the solution  $u$  can be written as

$u(x, y) = X(x) \cdot Y(y)$ , where  $X$  is a function of  $x$  alone and  $Y$  is a function of  $y$  alone. Substituting  $u$  and its partial derivatives in the given equation we get two ordinary differential equations. Solving that ordinary differential equation we get  $X$  and  $Y$ .

eg. ① Solve by the method of separation of variables

$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0. \quad \text{--- ①}$$

Here  $x$  and  $y$  are two independent variables and  $z$  is the dependent variable

ie  $z$  is a function of  $x$  and  $y$   
 $\therefore$  Assume the solution  $z = X Y$ , where  
 $X$  is a function of  $x$  alone and  $Y$  is  
a function of  $y$  alone

$$\therefore \frac{\partial z}{\partial x} = X' Y$$

$$\frac{\partial z}{\partial y} = X Y' \quad , \quad \frac{\partial^2 z}{\partial x^2} = X'' Y.$$

Substituting in (1)

$$X'' Y - 2 X' Y' + X Y'' = 0.$$

$$(X'' - 2 X') Y = -X Y'$$

$$\frac{X'' - 2 X'}{X} = \frac{-Y'}{Y} \quad \text{--- (2)}$$

{ Here L.H.S is a function of  $x$   
only and R.H.S is a function of  $y$  only.  
Since  $x$  and  $y$  are independent variables  
(2) is possible if each side is equal to  
a constant. }

$$\therefore \frac{X'' - 2 X'}{X} = \frac{-Y'}{Y} = a. \text{ (constant).}$$

$$\therefore \frac{X'' - 2 X'}{X} = a \Rightarrow X'' - 2 X' - a X = 0$$

$$\frac{d^2 X}{dx^2} - 2 \frac{dX}{dx} - a X = 0$$

$$(D^2 - 2D - a)X = 0 \quad (\text{o.d.e.}) \quad (2)$$

$$m^2 - 2m - a = 0 \quad (\text{A.E})$$

$$m = 1 \pm \sqrt{1+a}.$$

$$\therefore \text{C.F. } X = C_1 e^{(1+\sqrt{1+a})x} + C_2 e^{(1-\sqrt{1+a})x}.$$

$$(2) \Rightarrow \frac{-y'}{y} = a \Rightarrow \frac{-y'}{y} = a \Rightarrow \frac{y'}{y} + ay = 0$$

$$\frac{dy}{dy} + ay = 0.$$

$$(D+a)y = 0$$

$$\Rightarrow m+a=0$$

$$m = -a.$$

$$\therefore \text{C.F. } y = C_3 e^{-ay}.$$

$\therefore$  Solution of the given p.d.e is

$$Z = XY = \left[ C_1 e^{(1+\sqrt{1+a})x} + C_2 e^{(1-\sqrt{1+a})x} \right] C_3 e^{-ay}.$$

(2) Solve  $x \frac{\partial u}{\partial x} - 2y \frac{\partial u}{\partial y} = 0$ , by the method of separation of variables.

$$x \frac{\partial u}{\partial x} - 2y \frac{\partial u}{\partial y} = 0 \quad \text{--- (1)}$$

Here  $u$  as a function of  $x$  and  $y$

$\therefore u = xy$ , where  $x$  is a function of  $x$  only and  $y$  is a function of  $y$  only.

$$\frac{\partial u}{\partial x} = x'y, \quad \frac{\partial u}{\partial y} = xy'$$

Substituting in (1)

$$x x'y - 2y xy' = 0.$$

$$x x'y = 2y xy', \quad \text{separating the variables.}$$

$$\frac{x x'}{x} = \frac{2y y'}{y} = k \text{ (constant)}$$

$$\therefore \frac{x x'}{x} = k \Rightarrow x x' = kx$$

$$x \frac{dx}{dx} = kx.$$

$$\frac{dx}{x} = k \frac{dx}{x}.$$

$$\text{Integrating } \int \frac{dx}{x} = \int k \frac{dx}{x}$$

$$\log x = k \log x + \log a.$$

$$\log x = \log x^k + \log a \\ = \log (a x^k)$$

$$\therefore x = \underline{\underline{a x^k}}$$

Also  $\frac{2y y'}{y} = k \Rightarrow 2y y' = k y$

$$2y \frac{dy}{dy} = k y$$

$$\frac{dy}{y} = \frac{k dy}{2y}$$

Integrating  $\int \frac{dy}{y} = \frac{k}{2} \int \frac{dy}{y}$

$$\log y = \frac{k}{2} \log y + \log b.$$

$$= \log (y)^{k/2} + \log b.$$

$$= \log (y^{k/2} b)$$

$$\therefore y = b y^{k/2}$$

$$\therefore u = xy = (ax^k)(by^{k/2})$$

$$= c x^k y^{k/2} \quad (\because ab=c)$$

③ Solve  $3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$  where

$$u(x,0) = 4e^x.$$

Assume  $u = xy$ ,  $x$  is a function of  $x$  alone and  $y$  is a function of  $y$  only

$$\frac{\partial u}{\partial x} = x' y, \quad \frac{\partial u}{\partial y} = x y'$$

$$\textcircled{1} \Rightarrow 3x'y + 2xy' = 0$$

$$3x'y = -2xy'$$

$$\frac{3x'}{x} = -\frac{2y'}{y} = k.$$

$$\frac{3x'}{x} = k$$

$$3x' - xk = 0$$

$$3 \frac{dx}{dn} - kx = 0$$

$$(3D - k)x = 0 \text{ (o.d.e)}$$

$$3m - k = 0$$

$$m = \frac{k}{3}$$

$$\therefore x = C_1 e^{\frac{k}{3}n}.$$

$$-\frac{2y'}{y} = k.$$

$$-2y' = ky$$

$$-2 \frac{dy}{dy} = ky$$

$$(-2D - k)y = 0 \text{ (ode)}$$

$$-2m - k = 0$$

$$m = \frac{k}{-2}$$

$$\therefore y = C_2 e^{\left(\frac{k}{-2}\right)y}$$

$$\therefore u = xy$$

$$= C_1 e^{\frac{k}{3}n} \times C_2 e^{-\frac{k}{2}y}$$

$$= C e^{\frac{k}{3}n} e^{-\frac{k}{2}y} \text{ --- (2) } (\because C_1 C_2 = C)$$

$$\text{Given that } u(n,0) = 4e^{-n}.$$

$$\textcircled{2} \Rightarrow u(n,0) = C_1 e^{\frac{k}{3}n} e^0$$

$$= C e^{\frac{k}{3}n}$$

$$C e^{\frac{k}{3}n} = 4e^{-n}$$

$$\therefore \text{given } C e^{\frac{k}{3}n} = 4e^{-n}$$

$$\therefore C = 4 \text{ and } \frac{k}{3} = -1$$

$$k = -3$$

$$\therefore u = C e^{\frac{k}{3}x} e^{-\frac{k}{2}y}$$

$$= 4 e^{-x} e^{\frac{3}{2}y}$$

$$= \underline{\underline{4 e^{(-x + \frac{3}{2}y)}}}$$

(4)  $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$  gives  $u(0, y) = 8 e^{-3y}$ .

$$u = XY$$

$$\frac{\partial u}{\partial x} = X'Y$$

$$\frac{\partial u}{\partial y} = XY'$$

$$\therefore \frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y} \Rightarrow X'Y = 4XY'$$

$$\frac{X'}{X} = 4 \frac{Y'}{Y} = k \text{ (const)}$$

$$\frac{X'}{X} = k$$

$$X' - kX = 0$$

$$\frac{dx}{dx} - kx = 0 \text{ (o.d.e)}$$

$$(D - k)X = 0$$

$$m - k = 0$$

$$m = k$$

C.F.  $\therefore X = \underline{\underline{C_1 e^{kx}}}$

$$\frac{4Y'}{Y} = k$$

$$4Y' - kY = 0$$

$$4 \frac{dy}{dy} - kY = 0 \text{ (o.d.e)}$$

$$(4D - k)Y = 0$$

$$4m - k = 0$$

$$m = \frac{k}{4}$$

$\therefore Y = C_2 e^{\frac{k}{4}y}$

$$\therefore u = xy$$

$$= c_1 e^{kx} \times c_2 e^{\frac{k}{4}y}$$

Given that  $u(0, y) = 8e^{-3y}$

$$\therefore u(0, y) = c_1 e^0 c_2 e^{\frac{k}{4}y} = 8e^{-3y}$$

$$c_1 c_2 e^{\frac{k}{4}y} = 8e^{-3y}$$

$$\therefore c_1 c_2 = 8$$

$$\frac{k}{4} = -3 \Rightarrow k = \underline{\underline{-12}}$$

$$\therefore u = c_1 c_2 e^{kx} e^{\frac{k}{4}y}$$

$$= 8 e^{-12x} e^{-3y}$$

$$= 8 e^{-(12x+3y)}$$

⑤ Solve  $x^2 \frac{\partial^2 u}{\partial x^2} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$  — ①

$u = xy$  ,  $x$  is a function of  $x$  only  
 $y$  is a function of  $y$  only

$$\frac{\partial u}{\partial x} = x' y , \quad \frac{\partial u}{\partial y} = x y'$$

$$\textcircled{1} \Rightarrow x^2 x' y + y^2 x y' = 0$$

$x^2 x' y = -y^2 x y'$ , Separating the Variables

$$\frac{x^2 x'}{x} = \frac{-y^2 y'}{y} = k$$

$$\therefore \frac{x^2 x'}{x} = k$$

$$x^2 x' = kx$$

$$x^2 \frac{dx}{dx} = kx$$

$$\frac{dx}{x} = \frac{k dx}{x^2}$$

Integrating

$$\int \frac{dx}{x} = \int \frac{k dx}{x^2}$$

$$\log x = -\frac{k}{x} + a$$

$$\underline{\underline{x = e^{\left(-\frac{k}{x} + a\right)}}}$$

$$\frac{-y^2 y'}{y} = k$$

$$-y^2 y' = ky$$

$$-y^2 \frac{dy}{dy} = ky$$

$$\frac{dy}{y} = k \frac{dy}{-y^2}$$

Integrating

$$\log y = \frac{k}{y} + b$$

$$\therefore \underline{\underline{y = e^{\left(\frac{k}{y} + b\right)}}}$$

$\therefore$  Solution  $u = xy$

$$= e^{\left(-\frac{k}{x} + a\right)} \times e^{\left(\frac{k}{y} + b\right)}$$

$$= e^{-\frac{k}{x}} \cdot e^a \times e^{\frac{k}{y}} \cdot e^b$$

$$= e^{\frac{k}{y} - \frac{k}{x}} \cdot e^{(a+b)}$$

$$= C e^{k\left(\frac{1}{y} - \frac{1}{x}\right)}$$

H.W  
② Solve by the method of separation of variables.

$$\textcircled{1} \quad \frac{\partial u}{\partial x} - 2 \frac{\partial u}{\partial y} - u = 0, \quad u(x, 0) = b e^{-3x},$$

$$\textcircled{2} \quad \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} - 2u = 0$$

$$\textcircled{3} \quad u_x + u_y = 0.$$

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